Elements of AIML
Unit 02 – Lecture 02 Notes
Predicate Logic (First-Order Logic)

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1 Lecture Overview

This lecture introduces **predicate logic** (also called **first-order logic**, FOL). Compared to propositional logic, FOL can express:

- **objects** (Ali, UPES, Course-1),
- **properties** (Student(x)),
- **relations** between objects (Enrolled(x,y)),
- and **general rules** using quantifiers (∀, ∃).

Sources: Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 7–9 (Logical Agents, First-Order Logic, Inference).;Huth and Ryan, *Logic in Computer Science* (2e), selected sections on propositional and first-order logic.;Genesereth and Nilsson, *Logical Foundations of Artificial Intelligence*, selected chapters.

2 How to use these notes (self-study)

When translating English to FOL, write down (1) your domain, (2) your predicate meanings, and (3) the intended quantifier scope. Most errors in exams happen because one of these three was implicit in the student's mind but not written in logic. Try to translate each practice sentence *before* reading the provided answer.

3 Learning Outcomes

- Explain why propositional logic becomes verbose when there are many objects.
- Write basic FOL sentences with predicates, variables, and quantifiers.
- Interpret the meaning of universal and existential quantifiers.
- Translate simple English statements into FOL and avoid common pitfalls.

4 Keywords

- **Domain:** the set of objects that variables range over.
- **Constant:** a symbol naming a specific object (e.g., Ali).
- **Predicate:** a relation or property returning True/False (e.g., $Student(x)$).
- **Function:** maps objects to objects (e.g., $FatherOf(x)$).
- **Quantifier:** ∀ (for all), ∃ (there exists).
- **Scope:** the part of the sentence controlled by a quantifier.

5 Abbreviations and symbols

- **FOL**: First-Order Logic (predicate logic).
- **PL**: Propositional Logic.
- $\forall$: universal quantifier (for all).
- $\exists$: existential quantifier (there exists).
- $\neg, \wedge, \vee, \rightarrow$: NOT, AND, OR, IMPLIES.

6 Why predicate logic?

In propositional logic, we represent facts as atomic symbols. For example:

- p: "Ali is a student."
- q: "Sara is a student."

If we have 500 students, we need 500 separate symbols and cannot easily express a rule like:

"All students are hardworking."

Predicate logic solves this by introducing variables and quantifiers:

$$\forall x \; Student(x) \rightarrow WorksHard(x)$$

7 Syntax of FOL

7.1 Terms

Terms refer to objects:

- constants: Ali
- variables: x
- function applications: $FatherOf(x)$

7.2 Predicates vs functions (common confusion)

- A **predicate** returns True/False (it is a statement).
- A **function** returns an object (it is a term-producing operator).

Example: $Parent(Ali, Sara)$ could be a predicate (True/False). $FatherOf(Ali)$ is a function producing an object.

7.3 Atomic sentences

An atomic sentence is a predicate applied to terms, e.g.:

$$Student(Ali), \quad Enrolled(Ali, CS101), \quad Likes(x, AI)$$

7.4 Complex sentences

Build complex sentences using the same connectives as propositional logic:

$$\neg,\ \wedge,\ \vee,\ \rightarrow,\ \leftrightarrow$$

and include quantifiers:

$$\forall x\ (Student(x) \rightarrow WorksHard(x))$$

7.5 Free vs bound variables (scope)

- A variable is **bound** if it appears under a quantifier that declares it.
- A variable is **free** if it is not bound by any quantifier.

Example: in $\forall x\ Likes(x, AI)$, x is bound. In $Likes(x, AI)$ (alone), x is free and the sentence is not a complete statement unless we specify what x is.

8 Quantifiers and scope

8.1 Universal quantifier ($\forall$)

$\forall x\ P(x)$ means: for every object x in the domain, $P(x)$ holds.

8.2 Existential quantifier ($\exists$)

$\exists x\ P(x)$ means: there exists at least one object x such that $P(x)$ holds.

8.3 Scope and order

The **order** of quantifiers can change meaning.

Example. "Every student is enrolled in some course":

$$\forall x\ (Student(x) \rightarrow \exists y\ (Course(y) \wedge Enrolled(x, y)))$$

This allows different students to be enrolled in different courses.

"There exists a course that every student is enrolled in":

$$\exists y\ (Course(y) \wedge \forall x\ (Student(x) \rightarrow Enrolled(x, y)))$$

This asserts one shared course for all students.

8.4 Negating quantifiers (very useful later)

Quantifier negations:

$$\neg(\forall x\ P(x)) \equiv \exists x\ \neg P(x)$$

$$\neg(\exists x\ P(x)) \equiv \forall x\ \neg P(x)$$

These equivalences are important in clause form conversion and resolution (Unit II Lecture 03).

9 Semantics: interpreting FOL

To evaluate truth, we must choose:

- a **domain** (what objects exist),
- an **interpretation** (what constants, functions, and predicates mean).

Then sentences can be evaluated as True/False. A **model** is an interpretation where a set of sentences is true.

9.1 A tiny semantics example

Let the domain be the set $\{Ali, Sara\}$ and predicate $Student(\cdot)$. If the interpretation says:

$$Student(Ali) = T, \quad Student(Sara) = F,$$

then $\exists x\ Student(x)$ is true, while $\forall x\ Student(x)$ is false.

10 Translation: English → FOL

10.1 Practical workflow

1. Choose a domain (e.g., people, courses).
2. Define predicates/functions clearly (e.g., $Student(x)$, $Enrolled(x, y)$).
3. Identify quantifiers and their scope.
4. Build the final formula and check if it matches meaning.

10.2 Common patterns

- "All A are B": $\forall x\ A(x) \to B(x)$
- "Some A is B": $\exists x\ A(x) \land B(x)$
- "No A is B": $\forall x\ A(x) \to \neg B(x)$

10.3 More patterns (important in practice)

- "Only A are B": $\forall x\ B(x) \to A(x)$
- "At least one A": $\exists x\ A(x)$
- "At most one A" (sketch): $\forall x \forall y\ (A(x) \land A(y) \to x = y)$
- "Exactly one A" (sketch): (at least one) $\land$ (at most one)

10.4 Practice (with answers)

Assume domain: all people and courses in a university.

P1. "All humans are mortal." Answer: $\forall x\ Human(x) \to Mortal(x)$.

P2. "Some student likes AI." Answer: $\exists x\ Student(x) \land Likes(x, AI)$.

P3. "Every student has a mentor who is a faculty member." Answer:

$$\forall x\ (Student(x) \rightarrow \exists y\ (Faculty(y) \wedge MentorOf(y, x)))$$

P4. "No student is both absent and present." Answer: $\forall x\ Student(x) \rightarrow \neg(Absent(x) \wedge Present(x))$.

P5. "Only students can access the lab." Answer: $\forall x\ AccessLab(x) \rightarrow Student(x)$.

P6. "Exactly one student is class representative." Answer (one possible encoding):

$$\exists x\ (Student(x) \wedge Rep(x))\ \wedge\ \forall y\ ((Student(y) \wedge Rep(y)) \rightarrow y = x)$$

11 Bridge to next lecture: unification idea (preview)

Resolution in FOL needs **unification**, which is essentially "pattern matching" for terms. Example:

$$Likes(x, AI) \text{ matches } Likes(Ali, AI) \text{ with substitution } \{x/Ali\}.$$

We will study this formally in Lecture 03.

12 Practice (with answers)

P7. Translate and explain scope.

Translate: "Every student likes some course." **Answer:** $\forall x\ (Student(x) \rightarrow \exists y\ (Course(y) \wedge Likes(x, y)))$. This means each student may like a different course.

P8. Quantifier order changes meaning.

Translate: "There exists a course that every student likes." **Answer:** $\exists y\ (Course(y) \wedge \forall x\ (Student(x) \rightarrow Likes(x, y)))$. This asserts a single course liked by all students.

13 Common pitfalls (and how to avoid them)

- **Wrong implication direction:** "All A are B" is $A(x) \rightarrow B(x)$, not $B(x) \rightarrow A(x)$.
- **Missing scope parentheses:** always group the part controlled by a quantifier.
- **Quantifier order confusion:** $\forall x \exists y$ (depends on x) vs $\exists y \forall x$ (one y for all).
- **Forgetting the domain:** the truth of a FOL sentence depends on what objects x ranges over.
- **Free variables:** a sentence with a free variable is not a complete statement unless you quantify it.
- **Predicate vs function:** predicates are True/False statements; functions produce objects.

14 Exit questions (with answers)

Q1. Why is propositional logic insufficient for many knowledge bases?

Answer: It cannot represent objects and relations compactly; it requires one atom per fact and cannot express general rules using quantifiers.

Q2. Translate: "All humans are mortal" and "Some student likes AI".

Answer: $\forall x\ Human(x) \rightarrow Mortal(x)$ and $\exists x\ Student(x) \wedge Likes(x, AI)$.

Q3. Why does quantifier order matter?

Answer: $\forall x \exists y$ allows y to depend on x; $\exists y \forall x$ requires a single y that works for all x. The meaning changes.

Q4. Give one ambiguous English sentence without assumptions.

Answer: "Every student solved a problem." (same problem for all vs possibly different problems). Without clarifying, there are multiple valid FOL translations.

15 Summary

Predicate logic extends propositional logic by adding objects, relations, and quantifiers, enabling compact and expressive knowledge representation.

Sources: Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 7–9 (Logical Agents, First-Order Logic, Inference).;Huth and Ryan, *Logic in Computer Science* (2e), selected sections on propositional and first-order logic.;Genesereth and Nilsson, *Logical Foundations of Artificial Intelligence*, selected chapters.